Supplementary Material for: Flux blindness in neural solvers for heterogeneous subsurface flow: Spectral mechanisms, pre-training screening, and flux-informed learning

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S1. Reference Solver Verification and Spatial Grid Adequacy (128x128 vs. 256x256)

Prior to generating the 24-regime benchmark archive, the two-point flux approximation (TPFA) finite-volume reference solver (`src/subrel/solver.py`) passed four automated verification gates (`src/subrel/verify.py`):

1. **Method of Manufactured Solutions (MMS):** Using smooth analytical conductivity and head fields across grids $N \in \{16, 32, 64, 128\}$, the solver achieved empirical convergence orders of 2.00 in relative L_2 head error (1.64×10^{-3} at $N = 16$ down to 2.57×10^{-5} at $N = 128$) and 1.98 in interior relative L_2 Darcy-flux error (3.07×10^{-3} down to 4.98×10^{-5}).
2. **Homogeneous Linearity:** For uniform $K(\mathbf{x}) \equiv 1$, the solver recovered the exact linear ramp $h_0(x) = 1 - x$ with maximum pointwise error 8.77×10^{-15} and mean interior horizontal flux $\langle q_x \rangle = 1.000000$.
3. **Local Discrete Conservation:** Across heterogeneous realizations, interior cell-level discrete divergence satisfied $\max_{i,j} |\nabla_h \cdot \mathbf{F}_{i,j}| = 8.63 \times 10^{-12}$.
4. **Grid-Refinement Sensitivity (128^2 vs. 256^2):** Linearly interpolating individual $\ln K$ realizations onto a 256×256 grid, re-solving the TPFA system, and restricting fine face fluxes back to the 128×128 interfaces yielded relative face-flux differences of only 0.03% at $(\sigma^2, \lambda) = (0.5, 0.20)$, 1.3% at $(1.0, 0.03)$, 6.1% at $(4.0, 0.02)$, and 9.2% for the rough Matérn $\nu = 1/2$ field at $(2.0, 0.03)$—an order of magnitude smaller than the neural surrogate flux errors ($E_q = 1.44$–26.34). Similarly, in Pollock streamline particle tracking, solving on 256×256 preserved 100% reference particle arrival and

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shifted median travel time by only 1.45%; quadrupling injection density from 41 to 161 particles shifted fate percentages by < 2 percentage points.

S2. Vector-Norm Sensitivity and White-Noise Divergence Calibration

In the primary benchmark archive, the interior relative L_2 Darcy-flux error aggregates the two velocity components via the component-norm sum $E_q = (\|\delta q_x\|_2 + \|\delta q_y\|_2)/(\|q_x\|_2 + \|q_y\|_2)$. To verify invariance to vector-norm choice, all 3,264 test predictions were re-evaluated under the joint Euclidean vector norm:

$$E_{q,\mathrm{eucl}} = \frac{\sqrt{\|\delta q_x\|^2_{2,\Omega_{\mathrm{int}}} + \|\delta q_y\|^2_{2,\Omega_{\mathrm{int}}}}}{\sqrt{\|q_x\|^2_{2,\Omega_{\mathrm{int}}} + \|q_y\|^2_{2,\Omega_{\mathrm{int}}}}}, \tag{S1}$$

reproducing archived values to within 3.43×10^{-4} (`src/subrel/recompute_dual.py`). Replacing E_q with $E_{q,\mathrm{eucl}}$ changes binary failure labels (> 1.0) on only 5.2% of amortized evaluations (160/3,072 instances: 79 U-Net and 81 FNO), corresponding to overall failure rates of 86.3% versus 91.4% for U-Net and 70.0% versus 75.3% for FNO, while the joint conservative face-flux norm changes only 6.9% (224/3,264). All pre-training roughness screening rules remain identical.

In synthetic white-noise calibration experiments on the reference head field h_{ref}, the dimensionless local divergence imbalance $\mathcal{I}_{\mathrm{div}}$ scales linearly with uncorrelated cell-to-cell head-noise amplitude, taking values of 1.83 at 1% relative head noise, 9.16 at 5% noise, and 18.66 at 10% noise. Thus, U-Net's 24-regime median $\mathcal{I}_{\mathrm{div}} = 8.27$ corresponds to approximately 4.5% uncorrelated cell-to-cell head jitter.

S3. Capacity Ablations, Multi-Seed Reproducibility, and Prospective Blind Regimes

To decouple parameter count and spectral mode cutoff from loss-function alignment, we performed three controlled ablations across $(\sigma^2, \lambda) \in \{(0.5, 0.20), (1.0, 0.05), (2.0, 0.03)\}$:

1. **FNO Spectral Mode Expansion ($m \in \{12, 24, 32\}$):** Under head-only MSE training ($\alpha = 0$), increasing FNO retained modes from $m = 12$ to $m = 32$ reduces median E_q from 0.70 to 0.48 (-31%) at $(0.5, 0.20)$, from 1.44 to 1.18 (-18%) at $(1.0, 0.05)$, and from 4.21 to 3.60 (-14%) at $(2.0, 0.03)$, leaving the binary regime failure boundary unchanged.
2. **Matched-Capacity Parameter Controls:** Tripling U-Net parameters to match FNO (263,689 $\rightarrow$ 730,041 at width $w = 40$) reduces U-Net median E_q by 4.5%–16.0%, whereas halving FNO parameter elements to match U-Net (669,377 $\rightarrow$ 262,397 at width $w = 15$) increases FNO median E_q by 38%–64% while preserving each family's governing roughness invariant.
3. **Multi-Seed Reproducibility (4 Random Seeds):** Across four independent random initialization seeds (27 additional training runs) on $(0.5, 0.20)$, $(1.0, 0.05)$, and $(2.0, 0.03)$ for U-Net ($\alpha = 0$), head-only FNO ($\alpha = 0$), and flux-weighted FNO

($\alpha = 0.01$), binary regime classifications are 100% reproducible in 9/9 FNO configurations and 8/9 U-Net configurations (the sole threshold-straddling case being U-Net at $(0.5, 0.20)$, where seed medians are $0.84, 1.14, 0.94, 1.05$). At $(2.0, 0.03)$, Flux-FNO ($\alpha = 0.01$) achieves $E_q \in [1.41, 1.45]$ across all four seeds versus $[3.67, 4.21]$ for head-only FNO.

Table S1 reports the complete regime-by-regime results across all 12 prospective off-grid blind validation regimes (Blind Batch 1 and Blind Batch 2).

Table S1: Complete regime-level results across all 12 prospective off-grid blind validation regimes (64 test realizations per regime). Thresholds ($\sigma^2/\lambda \geq 3.33$ for U-Net; $\sigma^2 \geq 0.75$ for FNO) were frozen on the 24-regime atlas prior to generating or training these regimes.

Off-Grid Regime (σ^2, λ)	**Batch**	σ^2/λ	**U-Net Rule**	**U-Net** E_q	σ^2	**FNO Rule**	**FNO** E_q
$(0.75, 0.26)$ Smooth extrap.	Batch 1	2.88	Pass (≤ 1)	0.89 (Pass)	0.75	Flag (≥ 0.75)	0.71 (Pass)
$(0.75, 0.16)$ Interp.	Batch 1	4.69	Flag (> 1)	1.18 (Fail)	0.75	Flag (≥ 0.75)	0.77 (Pass)
$(0.75, 0.04)$ Interp.	Batch 1	18.75	Flag (> 1)	1.98 (Fail)	0.75	Flag (≥ 0.75)	1.04 (Fail)
$(1.50, 0.09)$ Interp.	Batch 1	16.67	Flag (> 1)	2.52 (Fail)	1.50	Flag (≥ 0.75)	1.57 (Fail)
$(1.50, 0.025)$ Interp.	Batch 1	60.00	Flag (> 1)	4.20 (Fail)	1.50	Flag (≥ 0.75)	2.83 (Fail)
$(3.00, 0.16)$ Interp.	Batch 1	18.75	Flag (> 1)	5.92 (Fail)	3.00	Flag (≥ 0.75)	3.54 (Fail)
$(3.00, 0.065)$ Interp.	Batch 1	46.15	Flag (> 1)	10.02 (Fail)	3.00	Flag (≥ 0.75)	5.69 (Fail)
$(6.00, 0.065)$ Rough extrap.	Batch 1	92.31	Flag (> 1)	82.31 (Fail)	6.00	Flag (≥ 0.75)	57.58 (Fail)
$(0.50, 0.10)$ Transition	Batch 2	5.00	Flag (> 1)	1.01 (Fail)	0.50	Pass (< 0.75)	0.73 (Pass)
$(0.75, 0.20)$ Transition	Batch 2	3.75	Flag (> 1)	1.03 (Fail)	0.75	Flag (≥ 0.75)	0.86 (Pass)
$(0.75, 0.12)$ Transition	Batch 2	6.25	Flag (> 1)	1.29 (Fail)	0.75	Flag (≥ 0.75)	0.79 (Pass)
$(1.00, 0.16)$ Transition	Batch 2	6.25	Flag (> 1)	1.59 (Fail)	1.00	Flag (≥ 0.75)	1.11 (Fail)

S4. Head-Error Spatial Phase-Locking and Empirical Covariance Orthogonality

In discrete operator form, $\delta\mathbf{q} = -B_K \delta\mathbf{h}$. Across 42 within-regime realization pairs and 21 cross-regime pairs, normalized head-error fields $\delta\mathbf{h}/\|\delta\mathbf{h}\|_2$ exhibit mean cosine similarities of 0.075 ± 0.254 and 0.038 ± 0.256, respectively, because pointwise head errors phase-lock to the specific high-∇Y interfaces of each realization. Consequently, replacing realization-specific error phases with a fixed empirical training covariance matrix $C_m = \mathbb{E}[\delta\mathbf{h}\,\delta\mathbf{h}^\top]$ inside $\mathrm{tr}(B_K C_m B_K^\top)$ yields low Spearman rank correlation with actual flux amplification ($\rho = 0.12$ for U-Net, 0.20 for FNO, and -0.20 for PINN). Similarly, the high-wavenumber spectral energy fraction of the reference head field h is non-predictive (instance-level ROC-AUC 0.39–0.58) because flow channeling smooths head fluctuations in high-variance media, confirming that pre-training screening must use medium-side roughness invariants of $K(\mathbf{x})$.